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И ВЫСШЕГО ОБРАЗОВАНИЯ
РОССИЙСКОЙ ФЕДЕРАЦИИ

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Преподаватели

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1 Цель работы

Разработать программу решения нелинейной одномерной краевой задачи методом конечных элементов. Провести сравнение метода простой итерации и метода Ньютона для решения данной задачи.

2 Задание

5. Уравнение: $-\text{div}(\lambda(u) \text{grad } u) + \sigma \frac{\partial u}{\partial t} = f$. Базисные функции – линейные.

3 Теоретическая часть

Уравнение:

$$-\text{div}(\lambda(u) \text{grad}(u)) + \sigma \partial u / \partial t = f,$$

С краевыми и начальными условиями:

$$u(S_1) = u_g$$

$$u(t_0) = u_0$$

Производную по времени на каждом временном слое t_s будем аппроксимировать по двухслойной неявной схеме:

$$\frac{\partial u}{\partial t} \Big|_{t=t_s} = \frac{u_s - u_{s-1}}{\Delta t_s}$$

Тогда решение задачи сводится к нахождению решения на каждом временном слое путем решения задачи:

$$-\text{div}(\lambda(u_s) \text{grad}(u_s)) + \frac{\sigma}{\Delta t_s} u_s = f_s + \frac{\sigma}{\Delta t_s} u_{s-1},$$

где $u_s = u(x, t_s)$ – решение на s-м временном слое, $\Delta t_s = t_s - t_{s-1}$ – шаг по времени, $f_s = f(x, t_s)$ – значение правой части на s-м временном слое.

В результате конечноэлементной аппроксимации нелинейной начально-краевой задачи получается система не линейных уравнений:

$$A(q_s)q_s = b(q_s).$$

Метод простой итерации:

Компоненты матрицы A и вектора правой части b определяются с помощью локальных матриц жесткости и массы:

$$\widehat{G} = \frac{\lambda_s^1(u(\widehat{q}_s)) + \lambda_s^2(u(\widehat{q}_s))}{2h_k} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \widehat{M}^t = \frac{\sigma h_k}{6\Delta t_s} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$
$$\widehat{b}^c = \frac{h_k}{6} \begin{bmatrix} f^k(\widehat{x}_1) \\ f^k(\widehat{x}_2) \end{bmatrix} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \widehat{b}^t = \frac{\sigma h_k}{6\Delta t_s} \begin{bmatrix} \widehat{q}_{s-1}^1 \\ \widehat{q}_{s-1}^2 \end{bmatrix} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \widehat{b} = \widehat{b}^c + \widehat{b}^t,$$

где $\lambda_s^i(u(q_s))$ – значение коэффициента лямбда в i узле, q_{s-1}^i – значения весов с прошлого временного слоя.

Решение будем находить с помощью метода просто итерации, где для каждого временного слоя будет брать начальное приближение (будем брать значение вектора весов с предыдущего слоя), для него собираться СЛАУ и находится новый вектор весов. Так будем продолжать пока не выполнится условие:

$$\frac{\|A(q^k)q^k - b(q^k)\|}{\|b\|} < \varepsilon,$$

где ε – маленькое число, выбранное как требуемая точность решения нелинейной задачи, или в случае если итерационный процесс “упрется” в максимальное число итераций.

Метод Ньютона:

Компоненты локальных линейаризованной матрицы и вектора правой части будут находится с помощью формул:

$$\hat{A}_{ij}^L = \hat{A}_{ij}(\hat{q}^0) + v \sum_{r=1}^{\hat{n}} \frac{\partial \hat{A}_{ir}(\hat{q})}{\partial \hat{q}_j} \Big|_{\hat{q}=\hat{q}^0} \hat{q}_r^0$$

$$\hat{b}_i^L = \hat{b}_i(\hat{q}^0) + v \sum_{j=1}^{\hat{n}} \hat{q}_j^0 \sum_{r=1}^{\hat{n}} \frac{\partial \hat{A}_{ir}(\hat{q})}{\partial \hat{q}_r} \Big|_{\hat{q}=\hat{q}^0} \hat{q}_r^0$$

$\hat{q}^0$ на первой итерации – начальное приближение, на последующих в качестве него берется вектор $\hat{q}^{k-1}$, v – коэффициент демпфирования.

Так как работаем с нелинейность только коэффициента $\lambda(u)$, то формула, по которой считается производная от $\lambda(u)$ на области Ω_m , имеет вид:

$$\frac{\partial \lambda_{\Omega_m}(u(\hat{q}, \hat{x}_l))}{\partial \hat{q}_j} \Big|_{\hat{q}=\hat{q}^0} = \frac{\partial \lambda_{\Omega_m}(u)}{\partial u} \Big|_{u=u(\hat{q}, \hat{x}_l)} \frac{\partial u(\hat{q}, \hat{x}_l)}{\partial \hat{q}_j} \Big|_{\hat{q}=\hat{q}^0} = \frac{\partial \lambda_{\Omega_m}(u)}{\partial u} \Big|_{u=u(\hat{q}, \hat{x}_l)} \hat{\psi}_j(\hat{x}_l)$$

Итерационный процесс строится аналогично тому, как был построен в методе простой итерации. Каждый шаг с помощью $\hat{q}^{k-1}$ с помощью вышеуказанных формул генерируется линейаризованная система и находится новое приближение $\hat{q}^k$. После чего проверяется критерий останова по относительной невязка. При оценке относительной невязки используется матрица и вектор нелинеаризованной системы.

4 Тестирование на работоспособность

4.1 Равномерная сетка

$$4.1.1 \ u(x, t) = x; \lambda(u) = 1; \sigma = 1; f(x, t) = 0; u(x, 0) = x$$

Метод простой итерации

Time layer 0: t = 0.00000, countNodes = 4, u* - u L2 = 2.3406e-16
Time layer 1: t = 1.50000, countNodes = 4, ht = 1.50000, countIteration = 0, u* - u L2 = 3.0638e-16
Time layer 2: t = 3.00000, countNodes = 4, ht = 1.50000, countIteration = 0, u* - u L2 = 3.0638e-16

Метод Ньютона

Time layer 0: t = 0.00000, countNodes = 4, u* - u L2 = 2.3406e-16
Time layer 1: t = 1.50000, countNodes = 4, ht = 1.50000, countIteration = 0, u* - u L2 = 4.8252e-16
Time layer 2: t = 3.00000, countNodes = 4, ht = 1.50000, countIteration = 0, u* - u L2 = 6.4562e-16

$$4.1.2 \ u(x, t) = x^2; \lambda(u) = 1; \sigma = 1; f(x, t) = -2; u(x, 0) = x^2$$

Метод простой итерации

Time layer 0: t = 0.00000, countNodes = 4, u* - u L2 = 4.6811e-16
Time layer 1: t = 1.50000, countNodes = 4, ht = 1.50000, countIteration = 0, u* - u L2 = 1.6868e-15
Time layer 2: t = 3.00000, countNodes = 4, ht = 1.50000, countIteration = 0, u* - u L2 = 1.6868e-15

Метод Ньютона

Time layer 0: t = 0.00000, countNodes = 4, u* - u L2 = 4.6811e-16
Time layer 1: t = 1.50000, countNodes = 4, ht = 1.50000, countIteration = 0, u* - u L2 = 9.4831e-16
Time layer 2: t = 3.00000, countNodes = 4, ht = 1.50000, countIteration = 0, u* - u L2 = 9.4831e-16

$$4.1.3 \ u(x, t) = t^2; \lambda(u) = 1; \sigma = 1; f(x, t) = 2t; u(x, 0) = 0$$

Метод простой итерации

Time layer 0: t = 0.00000, countNodes = 4, u* - u L2 = 0.0000e+00
Time layer 1: t = 1.50000, countNodes = 4, ht = 1.50000, countIteration = 0, u* - u L2 = 1.3295e+00
Time layer 2: t = 3.00000, countNodes = 4, ht = 1.50000, countIteration = 0, u* - u L2 = 1.8305e+00

Метод Ньютона

Time layer 0: t = 0.00000, countNodes = 4, u* - u L2 = 0.0000e+00
Time layer 1: t = 1.50000, countNodes = 4, ht = 1.50000, countIteration = 0, u* - u L2 = 1.3295e+00
Time layer 2: t = 3.00000, countNodes = 4, ht = 1.50000, countIteration = 0, u* - u L2 = 1.8305e+00

$$4.1.4 \ u(x, t) = x^2; \lambda(u) = u; \sigma = 1; f(x, t) = -6x^2; u(x, 0) = x^2$$

Метод простой итерации

Time layer 0: t = 0.00000, countNodes = 4, u* - u L2 = 4.6811e-16
Time layer 1: t = 1.50000, countNodes = 4, ht = 1.50000, countIteration = 0, u* - u L2 = 2.0314e-15
Time layer 2: t = 3.00000, countNodes = 4, ht = 1.50000, countIteration = 0, u* - u L2 = 3.4404e-15

Метод Ньютона

Time layer 0: t = 0.00000, countNodes = 4, u* - u L2 = 4.6811e-16
Time layer 1: t = 1.50000, countNodes = 4, ht = 1.50000, countIteration = 0, u* - u L2 = 9.7235e-16
Time layer 2: t = 3.00000, countNodes = 4, ht = 1.50000, countIteration = 0, u* - u L2 = 1.2579e-15

$$4.1.5 \ u(x, t) = x^2 + t; \lambda(u) = u; \sigma = 1; f(x, t) = 1 - 6x^2 - 2t; u(x, 0) = x^2$$

Метод простой итерации

Time layer 0: t = 0.00000, countNodes = 4, u* - u L2 = 4.6811e-16
Time layer 1: t = 1.50000, countNodes = 4, ht = 1.50000, countIteration = 11, u* - u L2 = 3.5684e-06
Time layer 2: t = 3.00000, countNodes = 4, ht = 1.50000, countIteration = 8, u* - u L2 = 4.6302e-06

Метод Ньютона

Time layer 0: t = 0.00000, countNodes = 4, u* - u L2 = 4.6811e-16
Time layer 1: t = 1.50000, countNodes = 4, ht = 1.50000, countIteration = 0, u* - u L2 = 6.1258e-16
Time layer 2: t = 3.00000, countNodes = 4, ht = 1.50000, countIteration = 0, u* - u L2 = 6.2882e-15

$$4.1.6 \ u(x, t) = x^2; \lambda(u) = 2u + 5; \sigma = 1; f(x, t) = -12x^2 - 10; u(x, 0) = x^2$$

Метод простой итерации

Time layer 0: t = 0.00000, countNodes = 4, u* - u L2 = 4.6811e-16
Time layer 1: t = 1.50000, countNodes = 4, ht = 1.50000, countIteration = 0, u* - u L2 = 2.1421e-15
Time layer 2: t = 3.00000, countNodes = 4, ht = 1.50000, countIteration = 0, u* - u L2 = 7.9438e-15

Метод Ньютона

Time layer 0: t = 0.00000, countNodes = 4, u* - u L2 = 4.6811e-16
Time layer 1: t = 1.50000, countNodes = 4, ht = 1.50000, countIteration = 0, u* - u L2 = 4.0147e-15
Time layer 2: t = 3.00000, countNodes = 4, ht = 1.50000, countIteration = 0, u* - u L2 = 8.9208e-16

$$4.1.7 \ (x, t) = x^2; \lambda(u) = \cos\left(\frac{u}{5}\right); \sigma = 1; f(x, t) = 0.8x^2 \sin\left(\frac{x^2}{5}\right) - 2\cos\left(\frac{x^2}{5}\right);$$

$$u(x, 0) = x^2$$

w = 1

Метод простой итерации

Time layer 0: t = 0.00000, countNodes = 4, u* - u L2 = 4.6811e-16
Time layer 1: t = 1.50000, countNodes = 4, ht = 1.50000, countIteration = 7, u* - u L2 = 2.4283e-02
Time layer 2: t = 3.00000, countNodes = 4, ht = 1.50000, countIteration = 6, u* - u L2 = 3.3951e-02

Метод Ньютона

Time layer 0: t = 0.00000, countNodes = 4, u* - u L2 = 4.6811e-16
Time layer 1: t = 1.50000, countNodes = 4, ht = 1.50000, countIteration = 2, u* - u L2 = 2.4277e-02
Time layer 2: t = 3.00000, countNodes = 4, ht = 1.50000, countIteration = 0, u* - u L2 = 3.3929e-02

w = 0.5

Метод простой итерации

Time layer 0: t = 0.00000, countNodes = 4, u* - u L2 = 4.6811e-16
Time layer 1: t = 1.50000, countNodes = 4, ht = 1.50000, countIteration = 9, u* - u L2 = 2.4280e-02
Time layer 2: t = 3.00000, countNodes = 4, ht = 1.50000, countIteration = 8, u* - u L2 = 3.3947e-02

Метод Ньютона

Time layer 0: t = 0.00000, countNodes = 4, u* - u L2 = 4.6811e-16
Time layer 1: t = 1.50000, countNodes = 4, ht = 1.50000, countIteration = 9, u* - u L2 = 2.4253e-02
Time layer 2: t = 3.00000, countNodes = 4, ht = 1.50000, countIteration = 7, u* - u L2 = 3.3896e-02

w = 0.1

Метод простой итерации

Time layer 0: t = 0.00000, countNodes = 4, u* - u L2 = 4.6811e-16
Time layer 1: t = 1.50000, countNodes = 4, ht = 1.50000, countIteration = 55, u* - u L2 = 2.4307e-02
Time layer 2: t = 3.00000, countNodes = 4, ht = 1.50000, countIteration = 49, u* - u L2 = 3.3978e-02

Метод Ньютона

Time layer 0: t = 0.00000, countNodes = 4, u* - u L2 = 4.6811e-16
Time layer 1: t = 1.50000, countNodes = 4, ht = 1.50000, countIteration = 57, u* - u L2 = 2.4223e-02
Time layer 2: t = 3.00000, countNodes = 4, ht = 1.50000, countIteration = 45, u* - u L2 = 3.3846e-02

w = 0.7

Метод простой итерации

Time layer 0: t = 0.00000, countNodes = 4, u* - u L2 = 4.6811e-16
Time layer 1: t = 1.50000, countNodes = 4, ht = 1.50000, countIteration = 6, u* - u L2 = 2.4271e-02
Time layer 2: t = 3.00000, countNodes = 4, ht = 1.50000, countIteration = 5, u* - u L2 = 3.3932e-02

Метод НьютонаTime layer 0: $t = 0.00000$, countNodes = 4, $\|u^* - u\|_{L2} = 4.6811e-16$ Time layer 1: $t = 1.50000$, countNodes = 4, $ht = 1.50000$, countIteration = 5, $\|u^* - u\|_{L2} = 2.4259e-02$ Time layer 2: $t = 3.00000$, countNodes = 4, $ht = 1.50000$, countIteration = 4, $\|u^* - u\|_{L2} = 3.3913e-02$ $w = 0.9$ **Метод простой итерации**Time layer 0: $t = 0.00000$, countNodes = 4, $\|u^* - u\|_{L2} = 4.6811e-16$ Time layer 1: $t = 1.50000$, countNodes = 4, $ht = 1.50000$, countIteration = 6, $\|u^* - u\|_{L2} = 2.4270e-02$ Time layer 2: $t = 3.00000$, countNodes = 4, $ht = 1.50000$, countIteration = 5, $\|u^* - u\|_{L2} = 3.3942e-02$ **Метод Ньютона**Time layer 0: $t = 0.00000$, countNodes = 4, $\|u^* - u\|_{L2} = 4.6811e-16$ Time layer 1: $t = 1.50000$, countNodes = 4, $ht = 1.50000$, countIteration = 3, $\|u^* - u\|_{L2} = 2.4275e-02$ Time layer 2: $t = 3.00000$, countNodes = 4, $ht = 1.50000$, countIteration = 3, $\|u^* - u\|_{L2} = 3.3942e-02$

$$4.1.8 \quad (x, t) = x^2 + t^2; \lambda(u) = \cos\left(\frac{u}{5}\right); \sigma = 1; f(x, t) = 2t - 2\cos\left(\frac{x^2 + t^2}{5}\right) + 0.8x^2 \sin\left(\frac{x^2 + t^2}{5}\right); u(x, 0) = x^2$$

Метод простой итерацииTime layer 0: $t = 0.0000$, countNodes = 4, $\|u^* - u\|_{L2} = 4.6811e-16$ Time layer 1: $t = 1.5000$, countNodes = 4, $ht = 1.50000$, countIteration = 10000, $\|u^* - u\|_{L2} = 1.8788e+00$ Time layer 2: $t = 3.0000$, countNodes = 4, $ht = 1.50000$, countIteration = 10000, $\|u^* - u\|_{L2} = 4.1296e+00$ **Метод Ньютона**Time layer 0: $t = 0.00000$, countNodes = 4, $\|u^* - u\|_{L2} = 4.6811e-16$ Time layer 1: $t = 1.50000$, countNodes = 4, $ht = 1.50000$, countIteration = 11, $\|u^* - u\|_{L2} = 1.0731e+01$ Time layer 2: $t = 3.00000$, countNodes = 4, $ht = 1.50000$, countIteration = 14, $\|u^* - u\|_{L2} = 1.4345e+02$ **4.2 Неравномерная сетка**

$$4.2.1 \quad u(x, t) = x^2; \lambda(u) = 1; \sigma = 1; f(x, t) = -2; u(x, 0) = x^2$$

Метод простой итерацииTime layer 0: $t = 0.00000$, countNodes = 8, $\|u^* - u\|_{L2} = 4.5059e-14$ Time layer 1: $t = 2.00000$, countNodes = 8, $ht = 2.00000$, countIteration = 0, $\|u^* - u\|_{L2} = 5.3088e-14$ Time layer 2: $t = 6.00000$, countNodes = 8, $ht = 4.00000$, countIteration = 0, $\|u^* - u\|_{L2} = 5.7472e-14$ Time layer 3: $t = 14.00000$, countNodes = 8, $ht = 8.00000$, countIteration = 0, $\|u^* - u\|_{L2} = 1.0323e-13$ **Метод Ньютона**Time layer 0: $t = 0.00000$, countNodes = 8, $\|u^* - u\|_{L2} = 4.5059e-14$ Time layer 1: $t = 2.00000$, countNodes = 8, $ht = 2.00000$, countIteration = 0, $\|u^* - u\|_{L2} = 6.3659e-14$ Time layer 2: $t = 6.00000$, countNodes = 8, $ht = 4.00000$, countIteration = 0, $\|u^* - u\|_{L2} = 7.1912e-14$ Time layer 3: $t = 14.00000$, countNodes = 8, $ht = 8.00000$, countIteration = 0, $\|u^* - u\|_{L2} = 9.6773e-14$

$$4.2.2 \quad u(x, t) = x^2; \lambda(u) = \cos(u); \sigma = 1; f(x, t) = 4x^2 \sin x^2 - 2\cos x^2; u(x, 0) = x^2$$

Метод простой итерацииTime layer0: $t = 0.00000$, countNodes = 4, $\|u^* - u\|_{L2} = 0.0000e+00$ Time layer1: $t = 2.00000$, countNodes = 4, $ht = 2.00000$, countIteration = 10000, $\|u^* - u\| = 3.2214e+02$ Time layer2: $t = 6.00000$, countNodes = 4, $ht = 4.00000$, countIteration = 10000, $\|u^* - u\| = 9.4697e+02$ Time layer3: $t = 14.00000$, countNodes = 4, $ht = 8.00000$, countIteration = 10000, $\|u^* - u\| = 2.6950e+03$ **Метод Ньютона**Time layer 0: $t = 0.00000$, countNodes = 4, $\|u^* - u\|_{L2} = 0.0000e+00$ Time layer 1: $t = 2.00000$, countNodes = 4, $ht = 2.00000$, countIteration = 10000, $\|u^* - u\| = 4.9030e+03$ Time layer 2: $t = 6.00000$, countNodes = 4, $ht = 4.00000$, countIteration = 10000, $\|u^* - u\| = 1.7586e+04$ Time layer 3: $t = 14.00000$, countNodes = 4, $ht = 8.00000$, countIteration = 10000, $\|u^* - u\| = 2.1066e+04$

5 Исследования

5.1 Равномерная сетка

5.1.1 Порядок аппроксимации и сходимость по x

$$u(x, t) = x^3; \lambda(u) = 1; \sigma = 1; f(x, t) = -6x; u(x, 0) = x^3; x \in [-1; 1], t = 0$$

h_x :

Метод простой итерации
Time layer 0: $t = 0.00000$, countNodes = 5, $\ u^* - u\ _{L2} = 5.1031e-03$
Метод Ньютона
Time layer 0: $t = 0.00000$, countNodes = 5, $\ u^* - u\ _{L2} = 5.1031e-03$

$\frac{h_x}{2}$:

Метод простой итерации
Time layer 0: $t = 0.00000$, countNodes = 9, $\ u^* - u\ _{L2} = 6.3789e-04$
Метод Ньютона
Time layer 0: $t = 0.00000$, countNodes = 9, $\ u^* - u\ _{L2} = 6.3789e-04$

$\frac{h_x}{4}$:

Метод простой итерации
Time layer 0: $t = 0.00000$, countNodes = 17, $\ u^* - u\ _{L2} = 7.9736e-05$
Метод Ньютона
Time layer 0: $t = 0.00000$, countNodes = 17, $\ u^* - u\ _{L2} = 7.9736e-05$

$$\log_2 \left(\frac{\|u(h_x) - u^*\|_{L_2}}{\|u(\frac{h_x}{2}) - u^*\|_{L_2}} \right) = \log_2 \left(\frac{5.10}{0.637} \right) \approx 3$$

$$\log_2 \left(\frac{\|u(\frac{h_x}{2}) - u^*\|_{L_2}}{\|u(\frac{h_x}{4}) - u^*\|_{L_2}} \right) = \log_2 \left(\frac{6.37}{0.797} \right) \approx 3$$

Порядок сходимости по x равен 3 как и в теории, порядок аппроксимации по x равен 2, т.к. по пространству используется квадратичная конечноэлементная аппроксимация(квадратичный базис).

т.к. $\lambda(u) = \text{const}$, то задача является линейной и матрицы для двух методов собираются идентично и в дальнейшем для $\lambda(u) = \text{const}$ будут рассматриваться только метод простой итерации.

5.1.2 Порядок аппроксимации и сходимость по t

$$u(x, t) = t^2; \lambda(u) = 1; \sigma = 1; f(x, t) = 2t; u(x, 0) = 0; x \in [-1; 1]$$

h_t :

Метод простой итерации
Time layer 0: $t = 0.00000$, countNodes = 5, $\ u^* - u\ _{L2} = 0.0000e+00$
Time layer 1: $t = 1.50000$, countNodes = 5, $h_t = 1.50000$, countIteration = 0, $\ u^* - u\ _{L2} = 6.1012e-01$

Time layer 2: t = 3.00000, countNodes = 5, ht = 1.50000, countIteration = 0, u* - u L2 = 7.3963e-01
--

$$\frac{h_t}{2};$$

Метод простой итерации

Time layer 0: t = 0.00000, countNodes = 5, u* - u L2 = 0.0000e+00
Time layer 1: t = 0.75000, countNodes = 5, ht = 0.75000, countIteration = 0, u* - u L2 = 2.5168e-01
Time layer 2: t = 1.50000, countNodes = 5, ht = 0.75000, countIteration = 0, u* - u L2 = 3.3974e-01
Time layer 3: t = 2.25000, countNodes = 5, ht = 0.75000, countIteration = 0, u* - u L2 = 3.7062e-01
Time layer 4: t = 3.00000, countNodes = 5, ht = 0.75000, countIteration = 0, u* - u L2 = 3.8145e-01

$$\frac{h_t}{4};$$

Метод простой итерации

Time layer 0: t = 0.00000, countNodes = 5, u* - u L2 = 0.0000e+00
Time layer 1: t = 0.37500, countNodes = 5, ht = 0.37500, countIteration = 0, u* - u L2 = 9.3257e-02
Time layer 2: t = 0.75000, countNodes = 5, ht = 0.37500, countIteration = 0, u* - u L2 = 1.4152e-01
Time layer 3: t = 1.12500, countNodes = 5, ht = 0.37500, countIteration = 0, u* - u L2 = 1.6658e-01
Time layer 4: t = 1.50000, countNodes = 5, ht = 0.37500, countIteration = 0, u* - u L2 = 1.7959e-01
Time layer 5: t = 1.87500, countNodes = 5, ht = 0.37500, countIteration = 0, u* - u L2 = 1.8635e-01
Time layer 6: t = 2.25000, countNodes = 5, ht = 0.37500, countIteration = 0, u* - u L2 = 1.8986e-01
Time layer 7: t = 2.62500, countNodes = 5, ht = 0.37500, countIteration = 0, u* - u L2 = 1.9168e-01
Time layer 8: t = 3.00000, countNodes = 5, ht = 0.37500, countIteration = 0, u* - u L2 = 1.9263e-01

$$\log_2 \left(\frac{\|u(h_t) - u^*\|_{L_2}}{\|u(\frac{h_t}{2}) - u^*\|_{L_2}} \right) = \log_2 \left(\frac{7.396}{3.815} \right) \approx 1$$

$$\log_2 \left(\frac{\|u(\frac{h_t}{2}) - u^*\|_{L_2}}{\|u(\frac{h_t}{4}) - u^*\|_{L_2}} \right) = \log_2 \left(\frac{3.815}{1.926} \right) \approx 1$$

Порядок сходимости по t равен 1, порядок аппроксимации по t равен 1, т.к. производная по времени аппроксимируется двухслойной неявной разностной схемой первого порядка.

5.2 Неравномерная сетка

5.2.1 Порядок аппроксимации и сходимость по x

$$u(x, t) = x^3; \lambda(u) = 1; \sigma = 1; f(x, t) = -6x; u(x, 0) = x^3; x \in [0; 14], t = 0, q = 2$$

$$h_{\max,1}: 7.46667$$

Метод простой итерации

Time layer 0: t = 0.00000, countNodes = 5, u* - u L2 = 3.6326e-02
--

$$h_{\max,2}: 7.02745$$

Метод простой итерации

Time layer 0: t = 0.00000, countNodes = 9, u* - u L2 = 2.9381e-02
--

$$h_{\max,3}: 7.00011$$

Метод простой итерации

Time layer 0: $t = 0.00000$, countNodes = 17, $\|u^* - u\|_{L2} = 2.8982e-02$

$$\frac{\ln\left(\frac{\|u(h_1) - u^*\|_{L_2}}{\|u(h_2) - u^*\|_{L_2}}\right)}{\ln(h_{\max,1}/h_{\max,2})} \approx 3.5$$

$$\frac{\ln\left(\frac{\|u(h_2) - u^*\|_{L_2}}{\|u(h_3) - u^*\|_{L_2}}\right)}{\ln(h_{\max,2}/h_{\max,3})} \approx 3.52$$

$$u(x, t) = x^3; \lambda(u) = 1; \sigma = 1; f(x, t) = -6x; u(x, 0) = x^3; x \in [-1; 1], t = 0, q = 1.3$$

$h_{\max,1}$: 0.63458

Метод простой итерации

Time layer 0: $t = 0.00000$, countNodes = 5, $\|u^* - u\|_{L2} = 9.5016e-03$

$h_{\max,2}$: 0.40098

Метод простой итерации

Time layer 0: $t = 0.00000$, countNodes = 9, $\|u^* - u\|_{L2} = 3.3237e-03$

$h_{\max,3}$: 0.23365

Метод простой итерации

Time layer 0: $t = 0.00000$, countNodes = 17, $\|u^* - u\|_{L2} = 2.2174e-03$

$$\frac{\ln\left(\frac{\|u(h_1) - u^*\|_{L_2}}{\|u(h_2) - u^*\|_{L_2}}\right)}{\ln(h_{\max,1}/h_{\max,2})} \approx 2.29$$

$$\frac{\ln\left(\frac{\|u(h_2) - u^*\|_{L_2}}{\|u(h_3) - u^*\|_{L_2}}\right)}{\ln(h_{\max,2}/h_{\max,3})} \approx 0.75$$

Для выбранной последовательности неравномерных сеток с параметром сгущения $q = 1.3/2$ экспериментальный порядок сходимости по пространственной переменной оказался нестабильным. Это указывает на чувствительность оценки к структуре неравномерной сетки и не позволяет надёжно определить порядок сходимости по данной последовательности сеток.

5.2.2 Порядок аппроксимации и сходимость по t

$$u(x, t) = t^2; \lambda(u) = 1; \sigma = 1; f(x, t) = 2t; u(x, 0) = 0; x \in [-1; 1],$$

$h_{\max,1}$: 1

Метод простой итерации

Time layer 0: $t = 0.00000$, countNodes = 5, $\|u^* - u\|_{L2} = 0.0000e+00$

Time layer 1: $t = 1.00000$, countNodes = 5, $ht = 1.00000$, countIteration = 0, $\|u^* - u\|_{L2} = 3.6774e-01$

Time layer 2: $t = 3.00000$, countNodes = 5, $ht = 2.00000$, countIteration = 0, $\|u^* - u\|_{L2} = 9.2090e-01$

$h_{\max,2}: 0.2$

Метод простой итерации

Time layer 0: $t = 0.00000$, countNodes = 5, $\|u^* - u\|_{L_2} = 0.0000e+00$
Time layer 1: $t = 0.20000$, countNodes = 5, $ht = 0.20000$, countIteration = 0, $\|u^* - u\|_{L_2} = 3.4265e-02$
Time layer 2: $t = 0.60000$, countNodes = 5, $ht = 0.40000$, countIteration = 0, $\|u^* - u\|_{L_2} = 1.1995e-01$
Time layer 3: $t = 1.40000$, countNodes = 5, $ht = 0.80000$, countIteration = 0, $\|u^* - u\|_{L_2} = 3.1467e-01$
Time layer 4: $t = 3.00000$, countNodes = 5, $ht = 1.60000$, countIteration = 0, $\|u^* - u\|_{L_2} = 7.2300e-01$

$h_{\max,3}: 0.01176$

Метод простой итерации

Time layer 0: $t = 0.00000$, countNodes = 5, $\|u^* - u\|_{L_2} = 0.0000e+00$
Time layer 1: $t = 0.01176$, countNodes = 5, $ht = 0.01176$, countIteration = 0, $\|u^* - u\|_{L_2} = 1.7885e-04$
Time layer 2: $t = 0.03529$, countNodes = 5, $ht = 0.02353$, countIteration = 0, $\|u^* - u\|_{L_2} = 8.5177e-04$
Time layer 3: $t = 0.08235$, countNodes = 5, $ht = 0.04706$, countIteration = 0, $\|u^* - u\|_{L_2} = 3.3203e-03$
Time layer 4: $t = 0.17647$, countNodes = 5, $ht = 0.09412$, countIteration = 0, $\|u^* - u\|_{L_2} = 1.1898e-02$
Time layer 5: $t = 0.36471$, countNodes = 5, $ht = 0.18824$, countIteration = 0, $\|u^* - u\|_{L_2} = 3.9017e-02$
Time layer 6: $t = 0.74118$, countNodes = 5, $ht = 0.37647$, countIteration = 0, $\|u^* - u\|_{L_2} = 1.1394e-01$
Time layer 7: $t = 1.49412$, countNodes = 5, $ht = 0.75294$, countIteration = 0, $\|u^* - u\|_{L_2} = 2.9276e-01$
Time layer 8: $t = 3.00000$, countNodes = 5, $ht = 1.50588$, countIteration = 0, $\|u^* - u\|_{L_2} = 6.7496e-01$

$$\frac{\ln\left(\frac{\|u(h_1) - u^*\|_{L_2}}{\|u(h_2) - u^*\|_{L_2}}\right)}{\ln(h_{\max,1}/h_{\max,2})} = \frac{\ln(0.92090/0.72300)}{\ln(2.0/1.6)} \approx 1.08$$

$$\frac{\ln\left(\frac{\|u(h_2) - u^*\|_{L_2}}{\|u(h_3) - u^*\|_{L_2}}\right)}{\ln(h_{\max,2}/h_{\max,3})} = \frac{\ln(0.72300/0.67496)}{\ln(1.6/1.505)} \approx 1.13$$

Порядок сходимости по t равен 1, порядок аппроксимации по t равен 1, т.к. производная по времени аппроксимируется двухслойной неявной разностной схемой первого порядка.

6 Выводы

На равномерной сетке метод Простой итерации и метод Ньютона показывают близкие результаты, за счёт простоты задач, но в задаче 4.1.4 Метод Ньютона оказался точнее, за счёт использования линеаризации системы, так же видно, что при разных числах правдоподобия решение сходится с разным количеством итераций. В задаче 4.1.7 для метода простой итерации $w = 0.9/0.8$, для метода Ньютона $w = 1$.

На не равномерной сетке получены удовлетворительные результаты. На задаче 4.2.2 решение получено с большой погрешностью за счёт того, что функция не представима в квадратичном базисе и сама функция стремительно изменяется на небольшом интервале.

7 Текст программы

7.3 Main.cpp

```
#include "Mesh.h"
#include "SimpleIteration.h"
#include "Newton.h"
#include <iostream>
#include <fstream>
#include <iomanip>
#include <sstream>
#include <string>
#include "LU.h"

std::string ToCommaString( double value, int precision, bool scientific = false )
{
    std::ostringstream stream;
    if ( scientific )
        stream << std::uppercase << std::scientific;
    else
        stream << std::fixed;

    stream << std::setprecision( precision ) << value;
    std::string text = stream.str( );
    for ( char& ch : text )
    {
        if ( ch == '.' )
            ch = ',';
    }
    return text;
}

template <typename Solver>
void WriteSolutionToFile( const Solver& solver, const std::string& fileName )
{
    std::ofstream file( fileName );
    if ( !file )
        return;

    file << "x\tq\tq*\tq - q*" << std::endl;

    Functions exactFunctions = solver.functions;

    for ( int i = 0; i < solver.n; i++ )
    {
        double x = 0.0;
        if ( i % 2 == 0 )
        {
```

```

        x = solver.mesh.meshX[i / 2];
    }
    else
    {
        int left = i / 2;
        x = ( solver.mesh.meshX[left] + solver.mesh.meshX[left + 1] ) / 2.0;
    }

    double exact = exactFunctions.u( x );
    double diff = solver.q[i] - exact;

    file
        << ToCommaString( x, 16 ) << "\t"
        << ToCommaString( solver.q[i], 16 ) << "\t"
        << ToCommaString( exact, 16 ) << "\t"
        << ToCommaString( diff, 2, true ) << std::endl;
    }
}

int main()
{
    int method, Id;

    std::cout << "Method (1 - Simple iteration, 2 - Newton): ";
    std::cin >> method;

    std::cout << "Function (1, 2, ...): ";
    std::cin >> Id;

    switch (method)
    {
    case 1:
    {
        SimpleIteration solver;
        solver.functions.id = Id;
        solver.useTimeDependent = true;

        solver.Input();
        solver.Solve();
        WriteSolutionToFile( solver, "out.txt" );

        break;
    }

    case 2:
    {
        Newton solver;
        solver.functions.id = Id;
        solver.useTimeDependent = true;

        solver.Input();
        solver.Solve();
        WriteSolutionToFile( solver, "out.txt" );

        break;
    }
    }

    return 0;
}

```

```

7.4 Basis.h
#pragma once
class Basis
{

```

```

public:
    double basis1( double x );
    double basis1Grad( double x );
    double basis2( double x );
    double basis2Grad( double x );
    double basis3( double x );
    double basis3Grad( double x );

    double **Quadrature( );
};

7.5 Basis.cpp
#include "Basis.h"
#include <cmath>

double Basis::basis1( double x )
{
    return 2 * ( x - 0.5 ) * ( x - 1 );
}

double Basis::basis1Grad( double x )
{
    return 4 * x - 3;
}

double Basis::basis2( double x )
{
    return -4 * x * ( x - 1 );
}

double Basis::basis2Grad( double x )
{
    return -4 * ( 2 * x - 1 );
}

double Basis::basis3( double x )
{
    return 2 * x * ( x - 0.5 );
}

double Basis::basis3Grad( double x )
{
    return 4 * x - 1;
}

double **Basis::Quadrature( )
{
    double **res = new double *[3];
    double t1 = -std::sqrt( 15.0 ) / 5.0;
    double t2 = 0.0;
    double t3 = +std::sqrt( 15.0 ) / 5.0;
    double w1 = 5.0 / 9.0;
    double w2 = 8.0 / 9.0;
    double w3 = 5.0 / 9.0;

    res[0] = new double[2] {( t1 + 1 ) / 2, 0.5 * w1};
    res[1] = new double[2] {( t2 + 1 ) / 2, 0.5 * w2};
    res[2] = new double[2] {( t3 + 1 ) / 2, 0.5 * w3};
    return res;
}

7.6 Mesh.h
#pragma once
#include <vector>
class Mesh

```

```

{
public:
    std::vector<double> meshX;
    std::vector<double> meshT;
    int n_x, n_t;

    bool uniformX = true;
    bool uniformT = true;

    void InputMeshes( );

    void CalculateUniformMesh( std::vector<double> &mesh, double a, double b, int n);
    void CalculateNonUniformMesh( std::vector<double> &mesh, double a, double b, int n,
double q );
};

```

7.7 Mesh.cpp

```

#include "Mesh.h"
#include <fstream>
#include <iostream>

void Mesh::InputMeshes( )
{
    double a_x, b_x, q_x, a_t, b_t, q_t;
    std::ifstream f( "mesh.txt" );
    if ( !f )
    {
        std::cerr << "File open error" << std::endl;
        return;
    }

    f >> uniformX >> n_x >> a_x >> b_x;
    meshX.resize(n_x);
    if ( uniformX )
        CalculateUniformMesh( meshX, a_x, b_x, n_x );
    else
    {
        f >> q_x;
        CalculateNonUniformMesh( meshX, a_x, b_x, n_x, q_x );
    }

    f >> uniformT >> n_t >> a_t >> b_t;
    meshT.resize( n_t );
    if ( uniformT )
        CalculateUniformMesh( meshT, a_t, b_t, n_t );
    else
    {
        f >> q_t;
        CalculateNonUniformMesh( meshT, a_t, b_t, n_t, q_t );
    }
}

void Mesh::CalculateUniformMesh( std::vector<double> &mesh, double a, double b, int n
)
{
    double h = ( b - a ) / ( n - 1 );
    for ( int i = 0; i < n; i++ )
        mesh[i] = a + i * h;
}

void Mesh::CalculateNonUniformMesh( std::vector<double> &mesh, double a, double b, int
n, double q )
{
    double h = ( b - a ) * ( 1 - q ) / ( 1 - pow( q, n - 1 ) );
    double hi = h;

```

```

    mesh[0] = a;
    double sum = 0;
    for ( int i = 1; i < n - 1; i++ )
    {
        sum += h * pow( q, i - 1 );
        mesh[i] = a + sum;
    }
    mesh[n-1] = b;
}

```

7.8 Func.h

```

#pragma once
#include <cmath>

class Functions
{
public:
    int id = 1;
    double currentTime = 0.0;

    double u(double x);
    double f(double x);
    double lambda(double u);
    double dlambda(double u);
    double sigma(double u);
    double sigmaTime(double t);
    double theta(double u);
    double beta(double u);
    double dbeta(double u);
    double uBeta(double u);
    double boundaryValue(int number, double x);
};

```

7.9 Func.cpp

```

#include "Func.h"
#include <numbers>

double Functions::u(double x)
{
    switch (id)
    {
    case 1:
        return x;

    case 2:
        return x * x;

    case 3:
        return currentTime * currentTime;

    case 4:
        return x * x;

    case 5:
        return x * x + currentTime;

    case 6:
        return x * x;

    default:
        return x;
    }
}

double Functions::f(double x)

```

```

{
    switch (id)
    {
    case 1:
        return 0.0;

    case 2:
        return -2.0;

    case 3:
        return 2.0 * currentTime;

    case 4:
        return -6.0 * x * x;

    case 5:
        return 1.0 - 6.0 * x * x - 2.0 * currentTime;

    case 6:
        return 4.0 * x * x * sin( x * x ) - 2.0 * cos( x * x );

    default:
        return 0.0;
    }
}

double Functions::lambda(double u)
{
    switch (id)
    {
    case 1:
    case 2:
    case 3:
        return 1.0;

    case 4:
        return u;
    case 5:
        return u;

    case 6:
        return cos( u );

    default:
        return 1.0;
    }
}

double Functions::dlambda(double u)
{
    switch (id)
    {
    case 1:
    case 2:
    case 3:
        return 0.0;

    case 4:
        return 1.0;
    case 5:
        return 1.0;

    case 6:
        return -sin( u );
    }
}

```

```

        default:
            return 0.0;
    }
}

double Functions::sigma(double u)
{
    switch (id)
    {
        default:
            return 1.0;
    }
}

double Functions::sigmaTime(double t)
{
    switch (id)
    {
        default:
            return 1.0;
    }
}

double Functions::theta(double u)
{
    switch (id)
    {
        default:
            return 0.0;
    }
}

double Functions::beta(double u)
{
    switch (id)
    {
        default:
            return 0.0;
    }
}

double Functions::dbeta(double u)
{
    switch (id)
    {
        default:
            return 0.0;
    }
}

double Functions::uBeta(double u)
{
    switch (id)
    {
        default:
            return 0.0;
    }
}

double Functions::boundaryValue(int number, double x)
{
    switch (number)
    {
        case 1:
            return u(x);
    }
}

```

```

        default:
            return 0.0;
    }
}

7.10 Matrix.h
#pragma once
#include <vector>
class Matrix
{
public:
    int n = 0;
    std::vector<double> di;
    std::vector<std::vector<double>> ggl;
    std::vector<std::vector<double>> ggu;
};

7.11 MatrixOperations.h
#pragma once
#include <vector>
#include "Matrix.h"
class MatrixOperations
{
public:
    Matrix MultMatScal( Matrix mat, double scal );
    std::vector<double> MultMatVec( Matrix mat, std::vector<double> vec );
    std::vector<double> MultVecScal( std::vector<double> vec, double scal );
    double DotProduct( std::vector<double> vec1, std::vector<double> vec2 );
    std::vector<double> AddVec( std::vector<double> vec1, std::vector<double> vec2 );
};

7.12 MatrixOperations.cpp
#include "MatrixOperations.h"

Matrix MatrixOperations::MultMatScal( Matrix mat, double scal )
{
    Matrix res = mat;
    for ( int i = 0; i < mat.n; i++ )
    {
        res.di[i] *= scal;
        for ( int j = 0; j < 2; j++ )
        {
            res.ggl[j][i] *= scal;
            res.ggu[j][i] *= scal;
        }
    }

    return res;
}

std::vector<double> MatrixOperations::MultMatVec( Matrix mat, std::vector<double> vec
)
{
    int n = mat.ggl[0].size();
    std::vector<double> y( mat.n, 0.0 );

    for ( int i = 0; i < mat.n; i++ )
    {
        // диагональ
        y[i] += mat.di[i] * vec[i];

        int maxI = std::max( n - i, 0 );
        // левая часть (j < i)
        for ( int j = maxI; j < n; j++ )
        {

```

```

        y[j - n + i] += mat.ggu[i][j] * vec[i];
    }

    // правая часть (j > i)
    for ( int j = maxI; j < n; j++ )
    {
        y[i] += mat.ggl[i][j] * vec[j - n + i];
    }
}

return y;
}

std::vector<double> MatrixOperations::MultVecScal( std::vector<double> vec, double
scal )
{
    std::vector<double> res = vec;
    for ( int i = 0; i < vec.size(); i++ )
        res[i] *= scal;
    return res;
}

double MatrixOperations::DotProduct( std::vector<double> vec1, std::vector<double>
vec2 )
{
    double res = 0;
    for ( int i = 0; i < vec1.size( ); i++ )
        res += vec1[i]*vec2[i];
    return res;
}

std::vector<double> MatrixOperations::AddVec( std::vector<double> vec1,
std::vector<double> vec2 )
{
    std::vector<double> res = vec1;
    for ( int i = 0; i < vec1.size( ); i++ )
        res[i] += vec2[i];
    return res;
}

```

7.13 LU.h

```

#pragma once
#include "Matrix.h"
#include <vector>
class LU
{
public:
    LU( ) {}
    LU( Matrix _matrix, std::vector<double> _b )
    {
        matrix = _matrix;
        b = _b;
        y = std::vector<double>( matrix.n, 0.0 );
        q = std::vector<double>( matrix.n, 0.0 );
    }
    Matrix matrix;
    std::vector<double> b;
    std::vector<double> y;
    std::vector<double> q;

    void calcLU( );

    void calcY( );

    void calcQ( );

```

```
};
```

7.14 LU.cpp

```
#include "LU.h"
void LU::calcLU( ) {
    auto &ggl = matrix.ggl, &ggg = matrix.ggu;
    auto &di = matrix.di;
    const int ASize = di.size( ), K = 2;

    for ( int i = 0; i < ASize; i++ ) {
        double sumDi = 0;
        int j = i - K;
        for ( int jl = 0; jl < K; jl++, j++ ) {
            double sum1 = 0, sum2 = 0;
            if ( j < 0 ) continue;
            int mj = i - j;
            for ( int mi = 0; mi < jl; mi++, mj++ ) {
                sum1 += ggu[i][mi] * ggl[j][mj];
                sum2 += ggu[j][mj] * ggl[i][mi];
            }
            ggu[i][jl] -= sum1;
            ggl[i][jl] = ( ggl[i][jl] - sum2 ) / di[j];
            sumDi += ggu[i][jl] * ggl[i][jl];
        }
        di[i] -= sumDi;
    }
}

void LU::calcY( ) {
    auto &ggl = matrix.ggl;
    const int ASize = b.size( ), K = 2;
    y.resize( ASize );

    for ( int i = 0; i < ASize; i++ ) {
        double sum = 0;
        int j = i - K;
        for ( int jl = 0; jl < K; jl++, j++ ) {
            if ( j < 0 ) continue;
            sum += ggl[i][jl] * y[j];
        }
        y[i] = b[i] - sum;
    }
}

void LU::calcQ( ) {
    auto &ggg = matrix.ggu;
    auto &di = matrix.di;
    const int ASize = y.size( ), K = 2;

    for ( int i = ASize - 1; i >= 0; i-- ) {
        q[i] = y[i] / di[i];
        int j = i - K;
        for ( int jl = 0; jl < K; jl++, j++ ) {
            if ( j < 0 ) continue;
            y[j] -= ggu[i][jl] * q[i];
        }
    }
    y.clear( );
}
```

7.15 Element.h

```
#pragma once
#include "Basis.h"
#include <vector>
```

```

#include "Func.h"

class Element
{
public:
    Functions localFunctions;
    Functions* functions = &localFunctions;

    Element( ) {}
    Element( std::vector<double> coords_, int cond_, std::vector<double> q_,
std::vector<double> _f, std::vector<double> _lambdaNodes, Functions* functions_ =
nullptr )
    {
        coords = coords_;
        cond = cond_;
        localB.resize( 3 );
        f = _f;
        lambdaNodes = _lambdaNodes;
        q = q_;
        SetFunctions( functions_ );
    }

    void SetFunctions( Functions* functions_ );

    //параметры задачи
    double lambdaApprox( double x );
    double materialLambda( double u );
    double materialDLambda(double u);
    std::vector<double> lambdaNodes;
    std::vector<double> f;

    double theta( double u );
    double beta( double u );
    double dbeta(double u);
    double uBeta( double u );
    double uh( double x );
    double uhPrevTime( double x );

    std::vector<double> coords;
    int cond = 0;
    std::vector<double> q;
    std::vector<double> qPrevTime;
    bool useTimeDependent = false;
    double deltaT = 1.0;
    double currentTime = 0.0;
    std::vector<std::vector<double>> localMatrix = std::vector<std::vector<double>>( 3,
std::vector<double>( 3, 0.0 ) );
    std::vector<double> localB;

    Basis basis;

    void BuildLocalMatrix( );
    void BuildLocalB( );
    void BuildLocalMatrixNewton(std::vector<double> q_0);
    void BuildLocalBNewton(std::vector<double> q_0);
};

7.16 Element.cpp
#include "Element.h"

void Element::SetFunctions( Functions* functions_ )
{
    functions = functions_ != nullptr ? functions_ : &localFunctions;
}

```

```

double Element::lambdaApprox( double x )
{
    return lambdaNodes[0] * basis.basis1( x ) + lambdaNodes[1] * basis.basis2( x ) +
    lambdaNodes[2] * basis.basis3( x );
}

double Element::materialLambda( double u )
{
    return functions->lambda( u );
}

double Element::materialDLambda( double u )
{
    return functions->dlambda( u );
}

double Element::theta( double u )
{
    return functions->theta( u );
}

double Element::beta( double u )
{
    return functions->beta( u );
}

double Element::dbeta( double u )
{
    return functions->dbeta( u );
}

double Element::uBeta( double u )
{
    return functions->uBeta( u );
}

double Element::uh( double x )
{
    return q[0] * basis.basis1( x ) + q[1] * basis.basis2( x ) + q[2] * basis.basis3( x );
}

double Element::uhPrevTime( double x )
{
    return qPrevTime[0] * basis.basis1( x ) + qPrevTime[1] * basis.basis2( x ) +
    qPrevTime[2] * basis.basis3( x );
}

void Element::BuildLocalMatrix( )
{
    localMatrix = std::vector<std::vector<double>>( 3, std::vector<double>( 3, 0.0 ) );
    double **quadrature = basis.Quadrature( );
    double jacobian = coords[1] - coords[0];
    for ( int i = 0; i < 3; i++ )
    {
        double wj = quadrature[i][1] * jacobian;
        double psi1 = basis.basis1( quadrature[i][0] );
        double psi2 = basis.basis2( quadrature[i][0] );
        double psi3 = basis.basis3( quadrature[i][0] );
        double gradPsi1 = basis.basis1Grad( quadrature[i][0] ) / jacobian;
        double gradPsi2 = basis.basis2Grad( quadrature[i][0] ) / jacobian;
        double gradPsi3 = basis.basis3Grad( quadrature[i][0] ) / jacobian;
        double sigmaVal = useTimeDependent ? 0.0 : functions->sigma( uh(
quadrature[i][0] ) );
        double lambdas = lambdaApprox( quadrature[i][0] );
    }
}

```

```

        localMatrix[0][0] += wj * ( lambdas * gradPsi1 * gradPsi1 + sigmaVal * psi1 *
psi1 + beta( uh( quadrature[i][0] ) ) * psi1 * psi1 );
        localMatrix[0][1] += wj * ( lambdas * gradPsi1 * gradPsi2 + sigmaVal * psi1 *
psi2 + beta( uh( quadrature[i][0] ) ) * psi1 * psi2 );
        localMatrix[0][2] += wj * ( lambdas * gradPsi1 * gradPsi3 + sigmaVal * psi1 *
psi3 + beta( uh( quadrature[i][0] ) ) * psi1 * psi3 );
        localMatrix[1][0] += wj * ( lambdas * gradPsi2 * gradPsi1 + sigmaVal * psi2 *
psi1 + beta( uh( quadrature[i][0] ) ) * psi2 * psi1 );
        localMatrix[1][1] += wj * ( lambdas * gradPsi2 * gradPsi2 + sigmaVal * psi2 *
psi2 + beta( uh( quadrature[i][0] ) ) * psi2 * psi2 );
        localMatrix[1][2] += wj * ( lambdas * gradPsi2 * gradPsi3 + sigmaVal * psi2 *
psi3 + beta( uh( quadrature[i][0] ) ) * psi2 * psi3 );
        localMatrix[2][0] += wj * ( lambdas * gradPsi3 * gradPsi1 + sigmaVal * psi3 *
psi1 + beta( uh( quadrature[i][0] ) ) * psi3 * psi1 );
        localMatrix[2][1] += wj * ( lambdas * gradPsi3 * gradPsi2 + sigmaVal * psi3 *
psi2 + beta( uh( quadrature[i][0] ) ) * psi3 * psi2 );
        localMatrix[2][2] += wj * ( lambdas * gradPsi3 * gradPsi3 + sigmaVal * psi3 *
psi3 + beta( uh( quadrature[i][0] ) ) * psi3 * psi3 );
        if ( useTimeDependent )
        {
            double sigmaTimeVal = functions->sigmaTime( currentTime ) / deltaT;
            localMatrix[0][0] += wj * sigmaTimeVal * psi1 * psi1;
            localMatrix[0][1] += wj * sigmaTimeVal * psi1 * psi2;
            localMatrix[0][2] += wj * sigmaTimeVal * psi1 * psi3;
            localMatrix[1][0] += wj * sigmaTimeVal * psi2 * psi1;
            localMatrix[1][1] += wj * sigmaTimeVal * psi2 * psi2;
            localMatrix[1][2] += wj * sigmaTimeVal * psi2 * psi3;
            localMatrix[2][0] += wj * sigmaTimeVal * psi3 * psi1;
            localMatrix[2][1] += wj * sigmaTimeVal * psi3 * psi2;
            localMatrix[2][2] += wj * sigmaTimeVal * psi3 * psi3;
        }
    }
}

void Element::BuildLocalB( )
{
    localB = std::vector<double>( 3, 0.0 );
    double **quadrature = basis.Quadrature( );
    double jacobian = coords[1] - coords[0];
    for ( int i = 0; i < 3; i++ )
    {
        double wj = quadrature[i][1] * jacobian;
        double psi1 = basis.basis1( quadrature[i][0] );
        double psi2 = basis.basis2( quadrature[i][0] );
        double psi3 = basis.basis3( quadrature[i][0] );
        double fVal = f[0] * psi1 + f[1] * psi2 + f[2] * psi3;
        localB[0] += wj * ( fVal * psi1 + theta( uh( quadrature[i][0] ) ) * psi1 + beta(
uh( quadrature[i][0] ) ) * uBeta( uh( quadrature[i][0] ) ) * psi1 );
        localB[1] += wj * ( fVal * psi2 + theta( uh( quadrature[i][0] ) ) * psi2 + beta(
uh( quadrature[i][0] ) ) * uBeta( uh( quadrature[i][0] ) ) * psi2 );
        localB[2] += wj * ( fVal * psi3 + theta( uh( quadrature[i][0] ) ) * psi3 + beta(
uh( quadrature[i][0] ) ) * uBeta( uh( quadrature[i][0] ) ) * psi3 );
        if ( useTimeDependent )
        {
            double sigmaTimeVal = functions->sigmaTime( currentTime ) / deltaT;
            double uhPrev = uhPrevTime( quadrature[i][0] );
            localB[0] += wj * sigmaTimeVal * uhPrev * psi1;
            localB[1] += wj * sigmaTimeVal * uhPrev * psi2;
            localB[2] += wj * sigmaTimeVal * uhPrev * psi3;
        }
    }
}

void Element::BuildLocalMatrixNewton( std::vector<double> q_0 )
{

```

```

localMatrix = std::vector<std::vector<double>>>( 3, std::vector<double>( 3, 0.0 ) );

double** quadrature = basis.Quadrature();
double jacobian = coords[1] - coords[0];
for ( int g = 0; g < 3; g++ )
{
    std::vector<double> psi( 3, 0.0 );
    std::vector<double> gradPsi( 3, 0.0 );
    double wj = quadrature[g][1] * jacobian;

    psi[0] = basis.basis1( quadrature[g][0] );
    psi[1] = basis.basis2( quadrature[g][0] );
    psi[2] = basis.basis3( quadrature[g][0] );

    gradPsi[0] = basis.basis1Grad( quadrature[g][0] ) / jacobian;
    gradPsi[1] = basis.basis2Grad( quadrature[g][0] ) / jacobian;
    gradPsi[2] = basis.basis3Grad( quadrature[g][0] ) / jacobian;

    double uhValue = uh( quadrature[g][0] );
    double sigmaVal = useTimeDependent ? 0.0 : functions->sigma( uhValue );
    double lambdaValue = lambdaApprox( quadrature[g][0] );
    double betaValue = beta( uhValue );
    double sigmaTimeValue = useTimeDependent ? functions->sigmaTime( currentTime ) /
deltaT : 0.0;

    double uhGradValue = 0.0;
    for ( int r = 0; r < 3; r++ )
    {
        uhGradValue += q_0[r] * gradPsi[r];
    }

    for ( int i = 0; i < 3; i++ )
    {
        for ( int j = 0; j < 3; j++ )
        {
            double A_ij = lambdaValue * gradPsi[i] * gradPsi[j] +
                sigmaVal * psi[i] * psi[j] +
                betaValue * psi[i] * psi[j] +
                sigmaTimeValue * psi[i] * psi[j];

            double dA_q = materialDLambda( q_0[j] ) * psi[j] * uhGradValue *
gradPsi[i];
            localMatrix[i][j] += wj * ( A_ij + dA_q );
        }
    }
}

void Element::BuildLocalBNewton( std::vector<double> q_0 )
{
    localB = std::vector<double>( 3, 0.0 );
    double** quadrature = basis.Quadrature();
    double jacobian = coords[1] - coords[0];
    for ( int g = 0; g < 3; g++ )
    {
        std::vector<double> psi( 3, 0.0 );
        std::vector<double> gradPsi( 3, 0.0 );
        double wj = quadrature[g][1] * jacobian;

        psi[0] = basis.basis1( quadrature[g][0] );
        psi[1] = basis.basis2( quadrature[g][0] );
        psi[2] = basis.basis3( quadrature[g][0] );

        gradPsi[0] = basis.basis1Grad( quadrature[g][0] ) / jacobian;
        gradPsi[1] = basis.basis2Grad( quadrature[g][0] ) / jacobian;

```

```

gradPsi[2] = basis.basis3Grad( quadrature[g][0] ) / jacobian;

double uhValue = uh( quadrature[g][0] );
double thetaValue = theta( uhValue );
double betaValue = beta( uhValue );
double uBetaValue = uBeta( uhValue );
double sigmaTimeValue = useTimeDependent ? functions->sigmaTime( currentTime ) /
deltaT : 0.0;
double uhPrev = useTimeDependent ? uhPrevTime( quadrature[g][0] ) : 0.0;
double fValue = f[0] * psi[0] + f[1] * psi[1] + f[2] * psi[2];

double uhGradValue = 0.0;
double dLambdaValue = 0.0;
for ( int r = 0; r < 3; r++ )
{
    uhGradValue += q_0[r] * gradPsi[r];
    dLambdaValue += materialDLambda( q_0[r] ) * q_0[r] * psi[r];
}

for ( int i = 0; i < 3; i++ )
{
    double B_i = fValue * psi[i] +
        thetaValue * psi[i] +
        betaValue * uBetaValue * psi[i] +
        sigmaTimeValue * uhPrev * psi[i];

    double dA_q = dLambdaValue * uhGradValue * gradPsi[i];
    localB[i] += wj * ( B_i + dA_q );
}
}
}

```

7.17 SimpleIteration.h

```

#pragma once
#include "Basis.h"
#include "Mesh.h";
#include "Element.h"
#include "Matrix.h"
#include "LU.h"
#include "MatrixOperations.h"
#include "Func.h"

class SimpleIteration
{
public:
    double eps = 1e-16;
    int maxIter = 100;
    int elemCount = 0;
    int n = 0;

    Functions functions;

    //Крайевые условия
    int boundaryConditions[2];
    double Condition(int number, double x );
    void Condition1( );

    std::vector<double> q;
    std::vector<double> q0Ld;
    std::vector<double> qPrevTime;

    //матрица
    Matrix matrix;

    //вектор правой части

```

```

std::vector<double> b;
std::vector<double> b0;

double deltaT;
bool useTimeDependent = false;
int timeLayer = 0;

double w = 0.01;

std::vector<Element> elements;
void Input( );
void UpdateLayerData( );
void BuildMatrix( );
void BuildB( );
void SolveIter( );
void Solve( );
double CalcResidual( );
double CalcL2Error( );
void PrintTimeLayerInfo( int layerIndex, int iterationCount );
std::vector<double> lambdaNodes;

Mesh mesh;
Basis basis;
LU lu;
MatrixOperations ops;
};

```

7.18 SimpleIteration.cpp

```

#include "SimpleIteration.h"
#include <iostream>
#include <fstream>
#include <iomanip>
#include <cmath>

double SimpleIteration::Condition( int number, double x )
{
    return functions.boundaryValue( number, x );
}

void SimpleIteration::Condition1( )
{
    if ( elements[0].cond == 1 )
    {
        matrix.di[0] = 1;
        b[0] = Condition( 1, 0 );
        for ( int i = 0; i < 2; i++ )
        {
            matrix.ggu[1][1] = 0;
            matrix.ggu[2][0] = 0;
        }
    }
    if ( elements[elemCount - 1].cond == 1 )
    {
        matrix.di[n - 1] = 1;
        b[n - 1] = Condition( 1, mesh.meshX[elemCount] );
        for ( int i = 0; i < 2; i++ )
        {
            matrix.ggl[n - 1][0] = 0;
            matrix.ggl[n - 1][1] = 0;
        }
    }
}

void SimpleIteration::Input( )
{

```

```

mesh.InputMeshes( );
std::ifstream file( "params.txt" );
if ( !file )
{
    std::cerr << "File open error" << std::endl;
    return;
}
file >> eps >> maxIter >> boundaryConditions[0] >> boundaryConditions[1];

elemCount = mesh.n_x - 1;
n = 3 * elemCount - elemCount + 1;
matrix.n = n;
matrix.ggl.resize( n );
matrix.ggu.resize( n );
for ( int i = 0; i < n; i++ )
{
    matrix.ggl[i] = std::vector<double>( 2, 0.0 );
    matrix.ggu[i] = std::vector<double>( 2, 0.0 );
}
matrix.di = std::vector<double>( n, 0.0 );

b.resize( n );
lambdaNodes.resize( n );
b0 = b;

std::vector<double> coords;
elements.resize( elemCount );
q = std::vector<double>( n, 0.0 );
if ( useTimeDependent )
{
    functions.currentTime = mesh.meshT[0];
    for ( int i = 0; i <= elemCount; i++ )
    {
        q[i * 2] = functions.u( mesh.meshX[i] );
        if ( i < elemCount )
            q[i * 2 + 1] = functions.u( ( mesh.meshX[i] + mesh.meshX[i + 1] ) / 2.0 );
    }
}
q[0] = Condition( 1, 0 );
q[n - 1] = Condition( 1, mesh.meshX[elemCount] );
qPrevTime = q;
std::vector<double> localB( 3, 0.0 );
std::vector<double> localLambdaNodes( 3, 0.0 );
for ( int i = 0; i < elemCount; i++ )
{
    coords = { mesh.meshX[i], mesh.meshX[i + 1] };
    for ( int j = 0; j < 3; j++ )
    {
        if ( i == 0 )
        {
            localB[j] = b[3 * i + j];
            localLambdaNodes[j] = lambdaNodes[3 * i + j];
        }
        else
        {
            localB[j] = b[2 * i + j];
            localLambdaNodes[j] = lambdaNodes[2 * i + j];
        }
    }
    elements[i] = Element( coords, 0, q, localB, localLambdaNodes, &functions );
}
elements[0].cond = boundaryConditions[0];
elements[elemCount - 1].cond = boundaryConditions[1];
UpdateLayerData( );

```

```

}

void SimpleIteration::BuildMatrix( )
{
    matrix.di = std::vector<double>( n, 0.0 );
    for ( int i = 0; i < n; i++ )
    {
        matrix.ggl[i] = std::vector<double>( 2, 0.0 );
        matrix.ggu[i] = std::vector<double>( 2, 0.0 );
    }

    for ( int i = 0; i < elemCount; i++ )
    {
        elements[i].useTimeDependent = useTimeDependent;
        elements[i].deltaT = deltaT;
        elements[i].currentTime = mesh.meshT[timeLayer];
        elements[i].BuildLocalMatrix( );
        for ( int j = 0; j < 3; j++ )
        {
            if ( i == 0 )
                matrix.di[3 * i + j] += elements[i].localMatrix[j][j];
            else
                matrix.di[2 * i + j] += elements[i].localMatrix[j][j];
        }
        matrix.ggl[2 * i + 1][1] += elements[i].localMatrix[1][0];
        matrix.ggl[2 * i + 2][0] += elements[i].localMatrix[2][0];
        matrix.ggl[2 * i + 2][1] += elements[i].localMatrix[2][1];

        matrix.ggu[2 * i + 1][1] += elements[i].localMatrix[0][1];
        matrix.ggu[2 * i + 2][0] += elements[i].localMatrix[0][2];
        matrix.ggu[2 * i + 2][1] += elements[i].localMatrix[1][2];
    }
}

void SimpleIteration::BuildB( )
{
    b = std::vector<double>( n, 0.0 );
    for ( int i = 0; i < elemCount; i++ )
    {
        elements[i].useTimeDependent = useTimeDependent;
        elements[i].deltaT = deltaT;
        elements[i].currentTime = mesh.meshT[timeLayer];
        elements[i].BuildLocalB( );
        for ( int j = 0; j < 3; j++ )
        {
            if ( i == 0 )
                b[3 * i + j] += elements[i].localB[j];
            else
                b[2 * i + j] += elements[i].localB[j];
        }
    }
}

void SimpleIteration::SolveIter( )
{
    qOld = q;
    UpdateLayerData( );
    BuildMatrix( );
    BuildB( );
    Condition1( );
    lu = LU( matrix, b );
    lu.calcLU( );
    lu.calcY( );
}

```

```

lu.calcQ( );
q = ops.AddVec( ops.MultVecScal( lu.q, w ), ops.MultVecScal( qOld, 1 - w ) );
std::vector<double> localQ( 3, 0.0 );
for ( int i = 0; i < elemCount; i++ )
{
    for ( int j = 0; j < 3; j++ )
    {
        if ( i == 0 )
        {
            localQ[j] = q[3 * i + j];
        }
        else
        {
            localQ[j] = q[2 * i + j];
        }
    }
    elements[i].q = localQ;
}
}

void SimpleIteration::Solve( )
{
    if ( !useTimeDependent )
    {
        int k = 0;
        SolveIter( );
        double residual = CalcResidual( );

        while ( residual > eps && k < maxIter )
        {
            k++;
            residual = CalcResidual( );
            std::cout << k << "\tresidual: " << residual << std::endl;
            for ( int i = 0; i < elemCount; i++ )
                for ( int j = 0; j < 3; j++ )
                {
                    if ( i == 0 )
                        elements[i].q[j] = q[3 * i + j];
                    else
                        elements[i].q[j] = q[2 * i + j];
                }
            SolveIter( );
        }

        return;
    }

    timeLayer = 0;
    functions.currentTime = mesh.meshT[0];
    PrintTimeLayerInfo( 0, 0 );

    for ( int s = 1; s < mesh.n_t; s++ )
    {
        timeLayer = s;
        deltaT = mesh.meshT[s] - mesh.meshT[s - 1];
        qPrevTime = q;
        q = qPrevTime;

        for ( int i = 0; i < elemCount; i++ )
        {
            std::vector<double> localQ( 3, 0.0 );
            std::vector<double> localPrevQ( 3, 0.0 );
            for ( int j = 0; j < 3; j++ )
            {

```

```

        if ( i == 0 )
        {
            localQ[j] = q[3 * i + j];
            localPrevQ[j] = qPrevTime[3 * i + j];
        }
        else
        {
            localQ[j] = q[2 * i + j];
            localPrevQ[j] = qPrevTime[2 * i + j];
        }
    }
    elements[i].q = localQ;
    elements[i].qPrevTime = localPrevQ;
}

int k = 0;
SolveIter( );
double residual = CalcResidual( );

while ( residual > eps && k < maxIter )
{
    k++;
    residual = CalcResidual( );
    SolveIter( );
}

functions.currentTime = mesh.meshT[s];
PrintTimeLayerInfo( s, k );
}
}

void SimpleIteration::UpdateLayerData( )
{
    functions.currentTime = mesh.meshT[timeLayer];
    b.assign( n, 0.0 );
    lambdaNodes.assign( n, 0.0 );

    for ( int i = 0; i <= elemCount; i++ )
    {
        b[i * 2] = functions.f( mesh.meshX[i] );
        lambdaNodes[i * 2] = functions.lambda( q[i * 2] );
        if ( i < elemCount )
        {
            double middle = ( mesh.meshX[i] + mesh.meshX[i + 1] ) / 2.0;
            b[i * 2 + 1] = functions.f( middle );
            lambdaNodes[i * 2 + 1] = functions.lambda( q[i * 2 + 1] );
        }
    }

    for ( int i = 0; i < elemCount; i++ )
    {
        for ( int j = 0; j < 3; j++ )
        {
            if ( i == 0 )
            {
                elements[i].f[j] = b[3 * i + j];
                elements[i].lambdaNodes[j] = lambdaNodes[3 * i + j];
            }
            else
            {
                elements[i].f[j] = b[2 * i + j];
                elements[i].lambdaNodes[j] = lambdaNodes[2 * i + j];
            }
        }
    }
}

```

```

}

double SimpleIteration::CalcResidual( )
{
    Matrix A_nonlin;
    A_nonlin.n = n;
    A_nonlin.di.assign( n, 0.0 );
    A_nonlin.ggl.assign( n, std::vector<double>( 2, 0.0 ) );
    A_nonlin.ggu.assign( n, std::vector<double>( 2, 0.0 ) );

    std::vector<double> b_nonlin( n, 0.0 );

    for ( int i = 0; i < elemCount; i++ )
    {
        elements[i].useTimeDependent = useTimeDependent;
        elements[i].deltaT = deltaT;
        elements[i].currentTime = mesh.meshT[timeLayer];
        elements[i].BuildLocalMatrix( );
        elements[i].BuildLocalB( );

        for ( int j = 0; j < 3; j++ )
        {
            if ( i == 0 )
                A_nonlin.di[3 * i + j] += elements[i].localMatrix[j][j];
            else
                A_nonlin.di[2 * i + j] += elements[i].localMatrix[j][j];
        }

        A_nonlin.ggl[2 * i + 1][1] += elements[i].localMatrix[1][0];
        A_nonlin.ggl[2 * i + 2][0] += elements[i].localMatrix[2][0];
        A_nonlin.ggl[2 * i + 2][1] += elements[i].localMatrix[2][1];

        A_nonlin.ggu[2 * i + 1][1] += elements[i].localMatrix[0][1];
        A_nonlin.ggu[2 * i + 2][0] += elements[i].localMatrix[0][2];
        A_nonlin.ggu[2 * i + 2][1] += elements[i].localMatrix[1][2];

        for ( int j = 0; j < 3; j++ )
        {
            if ( i == 0 )
                b_nonlin[3 * i + j] += elements[i].localB[j];
            else
                b_nonlin[2 * i + j] += elements[i].localB[j];
        }
    }

    if ( elements[0].cond == 1 )
    {
        A_nonlin.di[0] = 1;
        b_nonlin[0] = Condition( 1, 0 );
        A_nonlin.ggu[1][1] = 0;
        A_nonlin.ggu[2][0] = 0;
    }
    if ( elements[elemCount - 1].cond == 1 )
    {
        A_nonlin.di[n - 1] = 1;
        b_nonlin[n - 1] = Condition( 1, mesh.meshX[elemCount] );
        A_nonlin.ggl[n - 1][0] = 0;
        A_nonlin.ggl[n - 1][1] = 0;
    }

    std::vector<double> r = ops.AddVec(
        ops.MultMatVec( A_nonlin, q ),
        ops.MultVecScal( b_nonlin, -1 )
    );
}

```

```

    return ops.DotProduct( r, r ) / ops.DotProduct( b_nonlin, b_nonlin );
}

double SimpleIteration::CalcL2Error( )
{
    double sum = 0.0;
    for ( int i = 0; i < mesh.n_x; i++ )
    {
        double diff = q[2 * i] - functions.u( mesh.meshX[i] );
        sum += diff * diff;
    }
    return std::sqrt( sum );
}

void SimpleIteration::PrintTimeLayerInfo( int layerIndex, int iterationCount )
{
    double error = CalcL2Error( );
    std::streamsize oldPrecision = std::cout.precision();
    std::ios::fmtflags oldFlags = std::cout.flags();

    std::cout << std::fixed << std::setprecision( 5 )
        << "Time layer " << layerIndex << ": t = " << mesh.meshT[layerIndex]
        << ", countNodes = " << mesh.n_x;

    if ( layerIndex > 0 )
    {
        std::cout << ", ht = " << ( mesh.meshT[layerIndex] - mesh.meshT[layerIndex - 1]
    )
        << ", countIteration = " << iterationCount;
    }

    std::cout << std::scientific << std::setprecision( 4 )
        << ", ||u* - u||L2 = " << error << std::endl;

    std::cout.flags( oldFlags );
    std::cout.precision( oldPrecision );
}

```

7.19 Newton.h

```

#pragma once
#include "Basis.h"
#include "Mesh.h";
#include "Element.h"
#include "Matrix.h"
#include "LU.h"
#include "MatrixOperations.h"
#include "Func.h"

class Newton
{
public:
    double eps = 1e-16;
    int maxIter = 100;
    int elemCount = 0;
    int n = 0;

    Functions functions;

    //Краевые условия
    int boundaryConditions[2];
    double Condition(int number, double x);
    void Condition1();

    std::vector<double> q;
    std::vector<double> q0Ld;
}

```

```

std::vector<double> qPrevTime;

//матрица
Matrix matrix;

//вектор правой части
std::vector<double> b;
std::vector<double> b0;

double deltaT;
bool useTimeDependent = false;
int timeLayer = 0;

double w = 1.0;

std::vector<Element> elements;
void Input();
void UpdateLayerData();
void BuildMatrix(std::vector<double> qOld);
void BuildB(std::vector<double> qOld);
void SolveIter();
void Solve();
double CalcResidual();
double CalcL2Error();
void PrintTimeLayerInfo(int layerIndex, int iterationCount);
std::vector<double> lambdaNodes;

Mesh mesh;
Basis basis;
LU lu;
MatrixOperations ops;
};

```

7.20 Newton.cpp

```

#include "Newton.h"
#include <iostream>
#include <fstream>
#include <iomanip>
#include <cmath>

double Newton::Condition(int number, double x)
{
    return functions.boundaryValue(number, x);
}

void Newton::Condition1()
{
    if (elements[0].cond == 1)
    {
        matrix.di[0] = 1;
        b[0] = Condition(1, 0);
        for (int i = 0; i < 2; i++)
        {
            matrix.ggu[1][1] = 0;
            matrix.ggu[2][0] = 0;
        }
    }
    if (elements[elemCount - 1].cond == 1)
    {
        matrix.di[n - 1] = 1;
        b[n - 1] = Condition(1, mesh.meshX[elemCount]);
        for (int i = 0; i < 2; i++)
        {
            matrix.ggl[n - 1][0] = 0;
            matrix.ggl[n - 1][1] = 0;
        }
    }
}

```

```

    }
}

void Newton::Input()
{
    mesh.InputMeshes();
    std::ifstream file("params.txt");
    if (!file)
    {
        std::cerr << "File open error" << std::endl;
        return;
    }
    file >> eps >> maxIter >> boundaryConditions[0] >> boundaryConditions[1];

    elemCount = mesh.n_x - 1;
    n = 3 * elemCount - elemCount + 1;
    matrix.n = n;
    matrix.ggl.resize(n);
    matrix.ggu.resize(n);
    for (int i = 0; i < n; i++)
    {
        matrix.ggl[i] = std::vector<double>(2, 0.0);
        matrix.ggu[i] = std::vector<double>(2, 0.0);
    }
    matrix.di = std::vector<double>(n, 0.0);

    b.resize(n);
    lambdaNodes.resize(n);
    b0 = b;

    std::vector<double> coords;
    elements.resize(elemCount);
    q = std::vector<double>(n, 0.0);
    if (useTimeDependent)
    {
        functions.currentTime = mesh.meshT[0];
        for (int i = 0; i <= elemCount; i++)
        {
            q[i * 2] = functions.u(mesh.meshX[i]);
            if (i < elemCount)
                q[i * 2 + 1] = functions.u((mesh.meshX[i] + mesh.meshX[i + 1]) / 2.0);
        }
    }
    q[0] = Condition(1, 0);
    q[n - 1] = Condition(1, mesh.meshX[elemCount]);
    qPrevTime = q;
    std::vector<double> localB(3, 0.0);
    std::vector<double> localLambdaNodes(3, 0.0);
    for (int i = 0; i < elemCount; i++)
    {
        coords = { mesh.meshX[i], mesh.meshX[i + 1] };
        for (int j = 0; j < 3; j++)
        {
            if (i == 0)
            {
                localB[j] = b[3 * i + j];
                localLambdaNodes[j] = lambdaNodes[3 * i + j];
            }
            else
            {
                localB[j] = b[2 * i + j];
                localLambdaNodes[j] = lambdaNodes[2 * i + j];
            }
        }
    }
}

```

```

    }
    elements[i] = Element(coords, 0, q, localB, localLambdaNodes, &functions);
}
elements[0].cond = boundaryConditions[0];
elements[elemCount - 1].cond = boundaryConditions[1];
UpdateLayerData();

}

void Newton::BuildMatrix(std::vector<double> qOld)
{
    matrix.di = std::vector<double>(n, 0.0);
    for (int i = 0; i < n; i++)
    {
        matrix.ggl[i] = std::vector<double>(2, 0.0);
        matrix.ggu[i] = std::vector<double>(2, 0.0);
    }

    for (int i = 0; i < elemCount; i++)
    {
        std::vector<double> localQOld(3, 0.0);
        for (int j = 0; j < 3; j++)
        {
            if (i == 0)
                localQOld[j] = qOld[3 * i + j];
            else
                localQOld[j] = qOld[2 * i + j];
        }
        elements[i].useTimeDependent = useTimeDependent;
        elements[i].deltaT = deltaT;
        elements[i].currentTime = mesh.meshT[timeLayer];
        elements[i].BuildLocalMatrixNewton(localQOld);
        for (int j = 0; j < 3; j++)
        {
            if (i == 0)
                matrix.di[3 * i + j] += elements[i].localMatrix[j][j];
            else
                matrix.di[2 * i + j] += elements[i].localMatrix[j][j];
        }
        matrix.ggl[2 * i + 1][1] += elements[i].localMatrix[1][0];
        matrix.ggl[2 * i + 2][0] += elements[i].localMatrix[2][0];
        matrix.ggl[2 * i + 2][1] += elements[i].localMatrix[2][1];

        matrix.ggu[2 * i + 1][1] += elements[i].localMatrix[0][1];
        matrix.ggu[2 * i + 2][0] += elements[i].localMatrix[0][2];
        matrix.ggu[2 * i + 2][1] += elements[i].localMatrix[1][2];
    }
}

void Newton::BuildB(std::vector<double> qOld)
{
    b = std::vector<double>(n, 0.0);
    for (int i = 0; i < elemCount; i++)
    {
        std::vector<double> localQOld(3, 0.0);
        for (int j = 0; j < 3; j++)
        {
            if (i == 0)
                localQOld[j] = qOld[3 * i + j];
            else
                localQOld[j] = qOld[2 * i + j];
        }
        elements[i].useTimeDependent = useTimeDependent;
        elements[i].deltaT = deltaT;
    }
}

```

```

        elements[i].currentTime = mesh.meshT[timeLayer];
        elements[i].BuildLocalBNewton(localQOld);
        for (int j = 0; j < 3; j++)
        {
            if (i == 0)
                b[3 * i + j] += elements[i].localB[j];
            else
                b[2 * i + j] += elements[i].localB[j];
        }
    }
}

void Newton::SolveIter()
{
    qOld = q;
    UpdateLayerData();
    BuildMatrix(qOld);
    BuildB(qOld);
    Condition1();
    lu = LU(matrix, b);
    lu.calcLU();
    lu.calcY();
    lu.calcQ();
    q = ops.AddVec(ops.MultVecScal(lu.q, w), ops.MultVecScal(qOld, 1 - w));
    std::vector<double> localQ(3, 0.0);
    for (int i = 0; i < elemCount; i++)
    {
        for (int j = 0; j < 3; j++)
        {
            if (i == 0)
            {
                localQ[j] = q[3 * i + j];
            }
            else
            {
                localQ[j] = q[2 * i + j];
            }
        }
        elements[i].q = localQ;
    }
}

void Newton::Solve()
{
    if (!useTimeDependent)
    {
        int k = 0;
        SolveIter();
        double residual = CalcResidual();

        while (residual > eps && k < maxIter)
        {
            k++;
            residual = CalcResidual();
            std::cout << k << "\tresidual: " << residual << std::endl;
            for (int i = 0; i < elemCount; i++)
                for (int j = 0; j < 3; j++)
                {
                    if (i == 0)
                        elements[i].q[j] = q[3 * i + j];
                    else
                        elements[i].q[j] = q[2 * i + j];
                }
            SolveIter();
        }
    }
}

```

```

    }

    return;
}

timeLayer = 0;
functions.currentTime = mesh.meshT[0];
PrintTimeLayerInfo(0, 0);

for (int s = 1; s < mesh.n_t; s++)
{
    timeLayer = s;
    deltaT = mesh.meshT[s] - mesh.meshT[s - 1];
    qPrevTime = q;
    q = qPrevTime;

    for (int i = 0; i < elemCount; i++)
    {
        std::vector<double> localQ(3, 0.0);
        std::vector<double> localPrevQ(3, 0.0);
        for (int j = 0; j < 3; j++)
        {
            if (i == 0)
            {
                localQ[j] = q[3 * i + j];
                localPrevQ[j] = qPrevTime[3 * i + j];
            }
            else
            {
                localQ[j] = q[2 * i + j];
                localPrevQ[j] = qPrevTime[2 * i + j];
            }
        }
        elements[i].q = localQ;
        elements[i].qPrevTime = localPrevQ;
    }

    int k = 0;
    SolveIter();
    double residual = CalcResidual();

    while (residual > eps && k < maxIter)
    {
        k++;
        residual = CalcResidual();
        SolveIter();
    }

    functions.currentTime = mesh.meshT[s];
    PrintTimeLayerInfo(s, k);
}
}

void Newton::UpdateLayerData()
{
    functions.currentTime = mesh.meshT[timeLayer];
    b.assign(n, 0.0);
    lambdaNodes.assign(n, 0.0);

    for (int i = 0; i <= elemCount; i++)
    {
        b[i * 2] = functions.f(mesh.meshX[i]);
        lambdaNodes[i * 2] = functions.lambda(q[i * 2]);
        if (i < elemCount)
        {

```

```

        double middle = (mesh.meshX[i] + mesh.meshX[i + 1]) / 2.0;
        b[i * 2 + 1] = functions.f(middle);
        lambdaNodes[i * 2 + 1] = functions.lambda(q[i * 2 + 1]);
    }
}

for (int i = 0; i < elemCount; i++)
{
    for (int j = 0; j < 3; j++)
    {
        if (i == 0)
        {
            elements[i].f[j] = b[3 * i + j];
            elements[i].lambdaNodes[j] = lambdaNodes[3 * i + j];
        }
        else
        {
            elements[i].f[j] = b[2 * i + j];
            elements[i].lambdaNodes[j] = lambdaNodes[2 * i + j];
        }
    }
}

double Newton::CalcResidual()
{
    Matrix A_nonlin;
    A_nonlin.n = n;
    A_nonlin.di.assign(n, 0.0);
    A_nonlin.ggl.assign(n, std::vector<double>(2, 0.0));
    A_nonlin.ggu.assign(n, std::vector<double>(2, 0.0));

    std::vector<double> b_nonlin(n, 0.0);

    for (int i = 0; i < elemCount; i++)
    {
        elements[i].useTimeDependent = useTimeDependent;
        elements[i].deltaT = deltaT;
        elements[i].currentTime = mesh.meshT[timeLayer];
        elements[i].BuildLocalMatrix();
        elements[i].BuildLocalB();

        for (int j = 0; j < 3; j++)
        {
            if (i == 0)
                A_nonlin.di[3 * i + j] += elements[i].localMatrix[j][j];
            else
                A_nonlin.di[2 * i + j] += elements[i].localMatrix[j][j];
        }

        A_nonlin.ggl[2 * i + 1][1] += elements[i].localMatrix[1][0];
        A_nonlin.ggl[2 * i + 2][0] += elements[i].localMatrix[2][0];
        A_nonlin.ggl[2 * i + 2][1] += elements[i].localMatrix[2][1];

        A_nonlin.ggu[2 * i + 1][1] += elements[i].localMatrix[0][1];
        A_nonlin.ggu[2 * i + 2][0] += elements[i].localMatrix[0][2];
        A_nonlin.ggu[2 * i + 2][1] += elements[i].localMatrix[1][2];

        for (int j = 0; j < 3; j++)
        {
            if (i == 0)
                b_nonlin[3 * i + j] += elements[i].localB[j];
            else
                b_nonlin[2 * i + j] += elements[i].localB[j];
        }
    }
}

```

```

}

if (elements[0].cond == 1)
{
    A_nonlin.di[0] = 1;
    b_nonlin[0] = Condition(1, 0);
    A_nonlin.ggu[1][1] = 0;
    A_nonlin.ggu[2][0] = 0;
}
if (elements[elemCount - 1].cond == 1)
{
    A_nonlin.di[n - 1] = 1;
    b_nonlin[n - 1] = Condition(1, mesh.meshX[elemCount]);
    A_nonlin.ggl[n - 1][0] = 0;
    A_nonlin.ggl[n - 1][1] = 0;
}

std::vector<double> r = ops.AddVec(
    ops.MultMatVec(A_nonlin, q),
    ops.MultVecScal(b_nonlin, -1)
);

return ops.DotProduct(r, r) / ops.DotProduct(b_nonlin, b_nonlin);
}

double Newton::CalcL2Error()
{
    double sum = 0.0;
    for (int i = 0; i < mesh.n_x; i++)
    {
        double diff = q[2 * i] - functions.u(mesh.meshX[i]);
        sum += diff * diff;
    }
    return std::sqrt(sum);
}

void Newton::PrintTimeLayerInfo(int layerIndex, int iterationCount)
{
    double error = CalcL2Error();
    std::streamsize oldPrecision = std::cout.precision();
    std::ios::fmtflags oldFlags = std::cout.flags();

    std::cout << std::fixed << std::setprecision(5)
        << "Time layer " << layerIndex << ": t = " << mesh.meshT[layerIndex]
        << ", countNodes = " << mesh.n_x;

    if (layerIndex > 0)
    {
        std::cout << ", ht = " << (mesh.meshT[layerIndex] - mesh.meshT[layerIndex - 1])
            << ", countIteration = " << iterationCount;
    }

    std::cout << std::scientific << std::setprecision(4)
        << ", ||u* - u||L2 = " << error << std::endl;

    std::cout.flags(oldFlags);
    std::cout.precision(oldPrecision);
}

```